4	(a)	By reference to energy, explain what is meant by *electromotive force* (e.m.f.).	
		.....	
		.....	
		.....	[1]
	L1	e.m.f. is the **amount of energy transferred per unit charge from some other form of energy into electrical energy when charge is moved round a complete circuit.**	B1
	(b)	A 'potato cell' is formed by inserting a copper plate and a zinc plate into a potato. The circuit shown in Figure 4.1 is used in an investigation to determine the e.m.f. and internal resistance of the potato cell.	

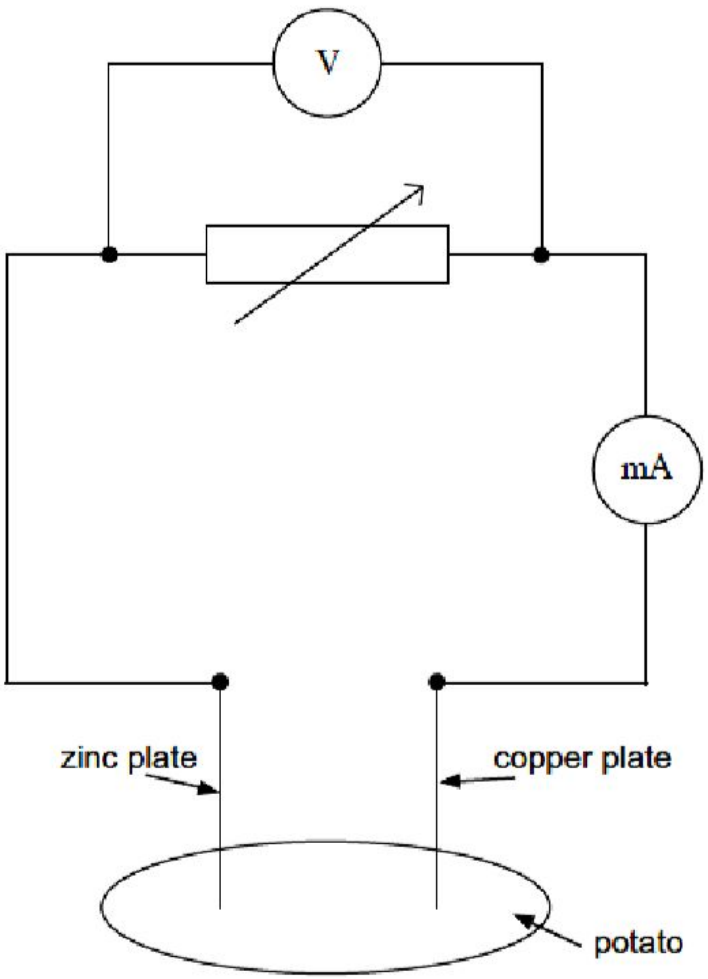


Figure 4.1

The plotted points × on Figure 4.2 show the data for current and voltage that were obtained by varying the resistance of the variable resistor. The line of best fit was then drawn through the plotted points.

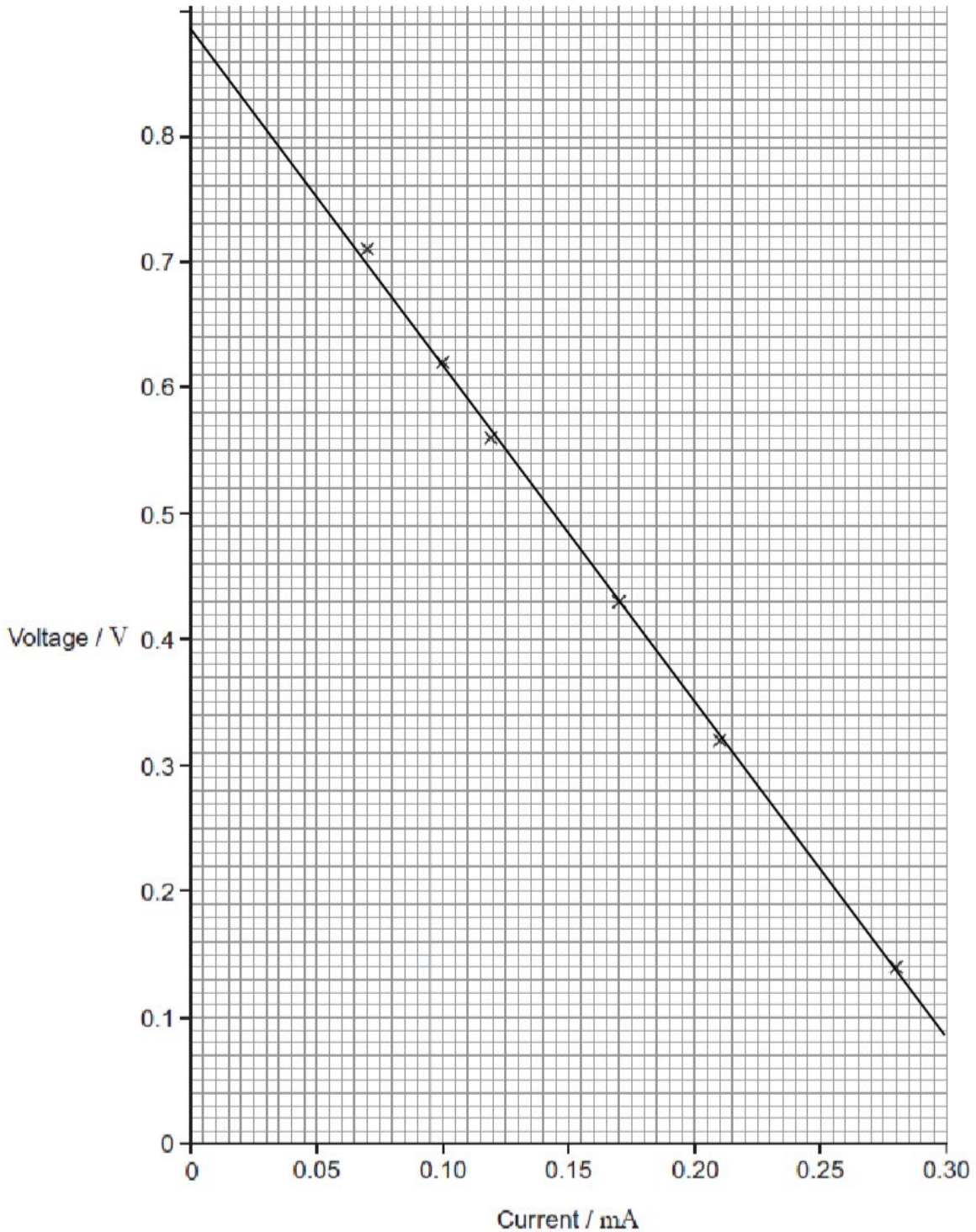
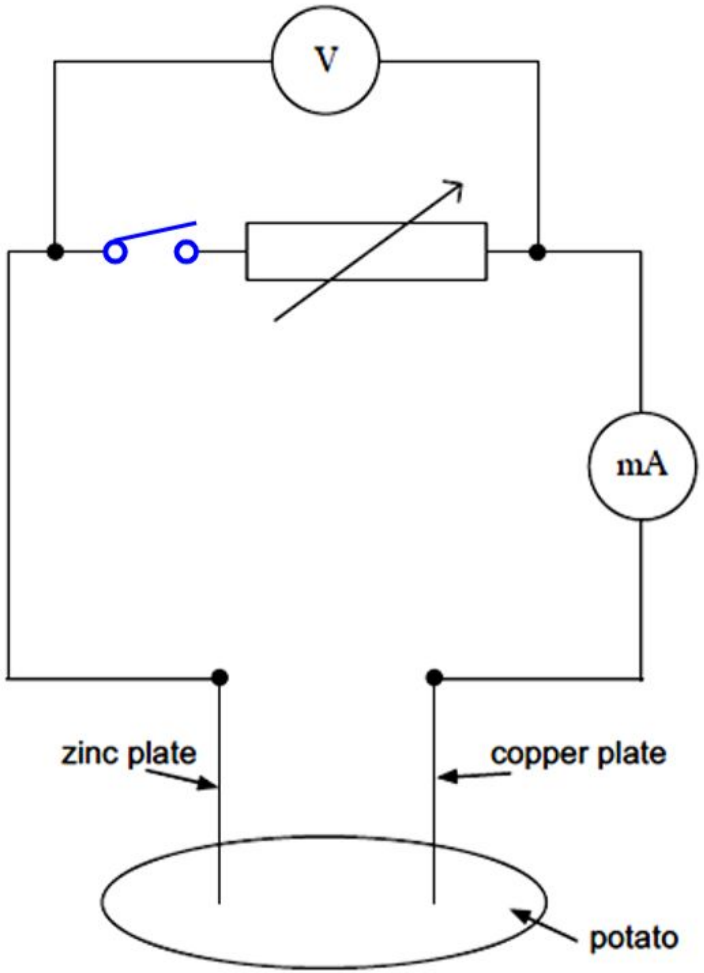


Figure 4.2

		(i)	State the e.m.f. of the potato cell.	
			e.m.f. = ...V	[1]
		L2	The graph in Figure 4.2 is described by the mathematical equation: $V = E - Ir$ When $I = 0$, $V = E$. Hence, the ***V*-intercept is the e.m.f.** → **e.m.f. = 0.885 V** **point must* ***be read off accurately to half the smallest division*** *on the graph, hence 0.885 V not 0.89 V.*	**B1**
		(ii)	Determine the internal resistance of the potato cell.	

			internal resistance = ...Ω	[3]
		L2	The graph in Figure 4.2 is described by the mathematical equation: $V = E - Ir$ **Internal resistance = – gradient** $= -\left[\frac{0.885-0.310}{(0-0.215)x10^{-3}}\right]$ = 2674.4 = **2670 Ω** ***C1 mark** – Evidence of equating internal resistance to negative of the gradient.* ***M1 mark** – Coordinates used must be separated by more than half the length of the line of best fit, and, correct reading off of coordinates. **If there is power of ten error, deduct under A1 mark, award M1 mark.* ***A1 mark** – Accept answers in the range 2600 – 2800 Ω, provided M1 mark scored.*	**C1** **M1** **A1**
		(iii)	Determine the voltage across the variable resistor when the power delivered to it is maximum. Explain your working.	
			voltage = ...V	[3]
		L2	**Power output is maximum when variable resistance *R* = internal resistance of the potato cell *r***	**B1**

			Using potential divider principle, **voltage across the variable resistance = 0.885 / 2** = 0.4425 **= 0.443 V**	**M1** **A1**
		(iv)	Explain why the voltages of the plotted points × on Figure 4.2 are always less than the e.m.f. of the potato cell.	
			.....	
			.....	
			.....	
			.....	
			.....	
			.....	
			.....	
			.....	
			.....	[3]
		L2	The **variable resistor's (maximum) resistance is not infinite. Current flows through the potato cell** regardless of how the variable resistor's resistance is varied **(OR, current is never zero), causing a** (non-zero) **potential difference exists across the internal resistance of the cell.** **Consequently, the terminal potential difference of the cell is always less than its e.m.f.** The **voltmeter reads the terminal potential difference** of the potato cell (not the e.m.f.).	**B1** **B1** **B1**
		(v)	In the space below, draw a circuit diagram to suggest a modification to the circuit in Figure 4.1 which allows the e.m.f. value of the potato cell to be read directly from the voltmeter.	

			[1]
	L2	**Solution:**  • **Add in a switch in series with the rheostat.** • **Switch must be in between the 2 terminals of the voltmeter.**	**B1**

[Total: 12]